

KIRAN BRAHMADEWARI

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EXPERIENCE

Software Engineer Intern

May 2021 – August 2021; June 2022 – September 2022;

Oracle Cloud Infrastructure (OCI)

May 2023 – August 2023; May 2024 – August 2024

- Built a distributed **Java** workflow to automate **Top of Rack (ToR) switch replacements**, resulting in an **80% decrease** in time spent triaging **Jira** tickets
- Created a command line tool in **Python** to automatically migrate **Terraform** alarm configurations into **Grafana** dashboards for real-time network monitoring
- Designed and implemented a **Java** backend using the **Data Access Object (DAO)** pattern to route fault code telemetry into a centralized **Oracle Database** and data warehouses
- Onboarded throttling microservices to **OCI** limits services using **Terraform**, internal tools, and scripts
- Wrote comprehensive documentation and automated test suites; achieved **90% method coverage**

Research Assistant

September 2023 – April 2024

Analytics for Cyber Defense Lab

- Developed client-server software in **Python** for large matrix multiplication outsourced to **Internet of Things (IoT)** devices, resulting in improved speed (i.e. faster runtime than **NumPy**'s `dot`, `matmult`, and `einsum` routines)
- Implemented **Multi-Party Computation (MPC)**, a **quantum-resistant cryptographic algorithm**

PROJECTS

AI Detection System for Real-Time Environmental Monitoring

January 2024 – May 2024

Capstone II

- Architected a cross-platform detection system using **Python**, **YOLOv8**, and **OpenCV** to identify harmful algae via livestream from **Internet of Things (IoT)** hardware (including **ESP32-CAM** and mobile devices)
- Implemented an **HTTP MJPEG streaming server** in **C++** and **C** for use with **Internet of Things (IoT)** hardware (including **ESP32-CAM** and mobile devices)
- Automated a **Machine Learning** pipeline in **Google Colab** to train algae detection models, achieving **89% mAP**, **82.6% precision**, and **89.4% recall**
- Led a 3-person engineering team through the full **Software Development Life Cycle (SDLC)** as **Tech Lead**; managed project requirements and wrote documentation

macOS Windows Translation and Automation Suite

March 2025 – Present

blog.lynkos.dev/posts/play-windows-games

- Built a **Bash** automation suite to run Windows applications on **Apple Silicon** using translation layers, including **Game Porting Toolkit (GPTk)**, **DXMT**, and **DXVK**
- Configured environment variables (e.g. `WINEPREFIX`, `WINEDLLOVERRIDES`, `MoltenVK` parameters, etc.) to optimize hardware and memory allocation on **Apple Silicon**
- **Optimized performance** with high-DPI scaling calculations, automated OS-level resource toggles, Windows registry tweaks, and more

Open Source Contributor — Chirpy Jekyll Theme

May 2026

PR #2642: fix: sidebar email link

- Fixed a critical navigation bug in a popular **Jekyll** theme (`jekyll-theme-chirpy`) on **GitHub** with **10.1k+ stars**, eliminating a Firefox-specific redirect failure that caused the sidebar email link to malfunction

SKILLS

Languages: Python, Java, Bash, Terraform, JavaScript, TypeScript, SQL, HTML, CSS, YAML, JSON, XML, Ruby

Frameworks: Agile, REST, React, JUnit, Mockito, Spring, Node.js, Expressif, Arduino, OpenCV, PyTorch, TensorFlow

Tools: Git, Linux, Docker, GitHub Actions, MongoDB, Grafana, Prometheus, CI/CD, Claude, Maven, Jira, Firebase

EDUCATION

Bachelor of Science in Computer Science

Graduated August 2025

Florida International University

- **Relevant Coursework:** Artificial Intelligence, Operating Systems, Net-Centric Computing, Data Structures, Discrete Mathematics, Systems Programming, Computer Architecture, Software Engineering, Algorithm Techniques, Principles of Programming Languages, Mobile Application Development, Fundamentals of Quantum Computing